TIP Visual Studio and the .NET CLI tools build your application automatically when you run it if they detect that a file has changed, so you generally won't need to perform this step explicitly yourself.

NuGet packages and the .NET CLI

One of the foundational components of .NET 7 cross-platform development is the .NET CLI, which provides several basic commands for creating, building, and running .NET 7 applications. Visual Studio effectively calls these commands automatically, but you can also invoke them directly from the command line if you're using a different editor. The most common commands used during development are

- `dotnet restore`
- `dotnet build`
- `dotnet run`

Each of these commands should be run inside your project folder and will act on that project alone. Except where explicitly noted, this is the case for all .NET CLI commands.

Most ASP.NET Core applications have dependencies on various external libraries, which are managed through the NuGet package manager. These dependencies are listed in the project, but the files of the libraries themselves aren't included. Before you can build and run your application, you need to ensure that there are local copies of each dependency on your machine. The first command, `dotnet restore`, ensures that your application's NuGet dependencies are downloaded and the files are referenced correctly by your project.

ASP.NET Core projects list their dependencies in the project's .csproj file, an XML file that lists each dependency as a `PackageReference` node. When you run `dotnet restore`, it uses this file to establish which NuGet packages to download. Any dependencies listed are available for use in your application.

The restore process typically happens implicitly when you build or run your application, as shown in the following figure, but it can be useful sometimes to run it explicitly, such as in continuous-integration build pipelines.